Xiaohua Cai

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EDUCATION

Master of Science in Computer Science, **Georgia Institute of Technology** GPA 4.0/4.0 December 2025
Master of Science in BioTechnology, **University of Alberta**, GPA 3.7/4.0 December 2018

SKILLS

Languages: Python, Java, JavaScript, PHP, TypeScript, SQL, HTML/CSS/SASS
Frameworks: React, Angular, Next.js, Node.js, Express, Laravel, Material UI, Tailwind CSS, Bootstrap
Database & Tools: PostgreSQL, MySQL, GraphQL, MongoDB, Docker, Git/GitHub, CI/CD, Postman, Netlify

WORK EXPERIENCE

Graduate Study & Parenting July 2024 - present

. Completed Georgia Institute of Technology's rigorous Master of Science in Computer Science program in December 2025 with a perfect 4.0/4.0 GPA.
. Career break for parenting
. Continued developing technical skills through hands-on practice in algorithms, data structures, system design, and software engineering.

Full Stack Developer December 2022 - June 2024

Affinity Bridge Inc.

. Built and shipped an internal full-stack application using Next.js, TypeScript and Prisma; integrated NextAuth for authentication and developed REST API endpoints and relational database operations.
. Developed React frontend features and PHP-based RESTful APIs for the NovaSafe application; wrote and optimized SQL queries supporting CRUD operations and application data management.
. Built the Environmental Conventions Index (ECI) with Next.js: (live site).
  . Developed server- and client-rendered pages.
  . Designed and implemented interactive geographic maps using React Simple Maps to visualize environmental data.
  . Developed responsive, interactive scatter and line charts with ECharts to visualize environmental trends and country-level progress.
. Delivered React frontend features across multiple client applications, building reusable responsive components and integrating application interfaces with backend APIs.

Full Stack Developer January 2022 - December 2022

AMI RockChain Inc.

. Built front end features using Angular, TypeScript and Nebular UI library according to the Figma design
. Created RESTful APIs in PHP Laravel Framework and implemented CRUD operations to read, create, update and delete data
. Generated migration and set up relational data models
. Built queries to retrieve data from PostgreSQL database
. Adapted Model-View-Controller (MVC) pattern to develop web applications
. Built web mapping applications using Leaflet library and Google map API
. Managed subscribed members using MailChimp API
. Developed a graph tool using Echarts Library for user entering data into table to automatically generate charts

Full Stack Developer (Contract) September 2021 - January 2022

Explorer Hop

Project 1: *Internal used app*

. Built front end features using React.js, Material UI, TypeScript
. Built RESTful API and backend logics with Node.js Express, PostgreSQL
. Worked closely with the designer to implement web app user interfaces

Project 2: *Shopify App*

. Built Shopify admin app frontend features using React.js, Shopify App Bridge and Polaris components
. Implemented queries to update, retrieve data from Shopify GraphQL API
. Implemented webhooks into app to notify the app when user create a product order

Research and Development Scientist January 2019- September 2021

SBI BioEnergy Inc.

. Developed VBA automation tools to calculate mass balances for 10+ lab-scale reactors, reducing repetitive manual calculations and supporting experimental analysis.
. Utilized process-control software to automate and monitor chemical reaction processes across laboratory experiments.
. Analyzed, processed, and visualized experimental data, and regularly presented findings to process and catalyst development teams.